Introduction

This work is grounded in the processing and interpretation of in-cylinder pressure, one of the most informative signals for characterising reciprocating engines.

In the first part, key combustion parameters are derived for a detailed analysis of the combustion evolution. The second part extends the role of the same pressure signal to a different objective: estimating the trapped air mass, which is essential for engine control and efficiency optimisation. The two sections are therefore complementary, relying on the same measurement framework while applying it to distinct engineering challenges.

1 Advanced combustion analysis – Part I

Combustion performances of a spark ignition engine with the following specification have been investigated using MATLAB.

Description	Value	Unit
B (bore)	77	*mm*
s (stroke)	85,8	*mm*
l (connecting rod)	138,4	*mm*
r (compression ratio)	10,5	-
R= 2l/s (conrod to stroke ratio)	3,2261	-
V_d (displacement volume)	399,5	*cm^3*
$V_{clearance}$	41,12	*cm^3*
n_{cyl}	4	-

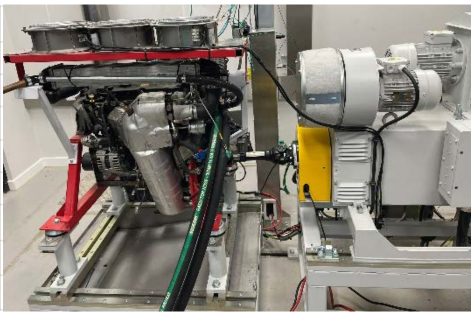


Table 1: BMW Mini engine specification and test bed illustration

Mass fraction burned (*Mfb)*, heat release rate (*HRR),* and cumulative heat release(Q_{cum}) are derived from one cycle measured cylinder pressure, allowing to perform comparisons between the following operating conditions:

Conditions	**LOAD** *[Nm]*	**SPEED** *[rpm]*	θ_{SOC} *[CA°]*	θ_{EOC} *[CA°]*
Fixed speed (2400 rpm)	40	-	-18	23
	60		-16	24
	90		-14	28
Fixed load (90 Nm)	-	1600	-9	29
		2000	-12	29
		2400	-14	28

Table 2: Engine conditions tested and evaluated combustion intervals

To define the start and the end of combustion, the pV^n diagram was investigated using a semi-logarithmic scale. Start of combustion (SOC) is identified by the departure of the curve from the straight line before TDC, as shown on the left side of Figure 1. Being the end of combustion (EOC) more difficult to evaluate graphically with respect to the SOC, it was identified as the crank angle after TDC at which the pV^n slope drops below a certain threshold (defined as 5% of its peak value), as the curve returns to a constant trend. This behaviour marks the point at which most of the fuel has been burned, and combustion is essentially complete.

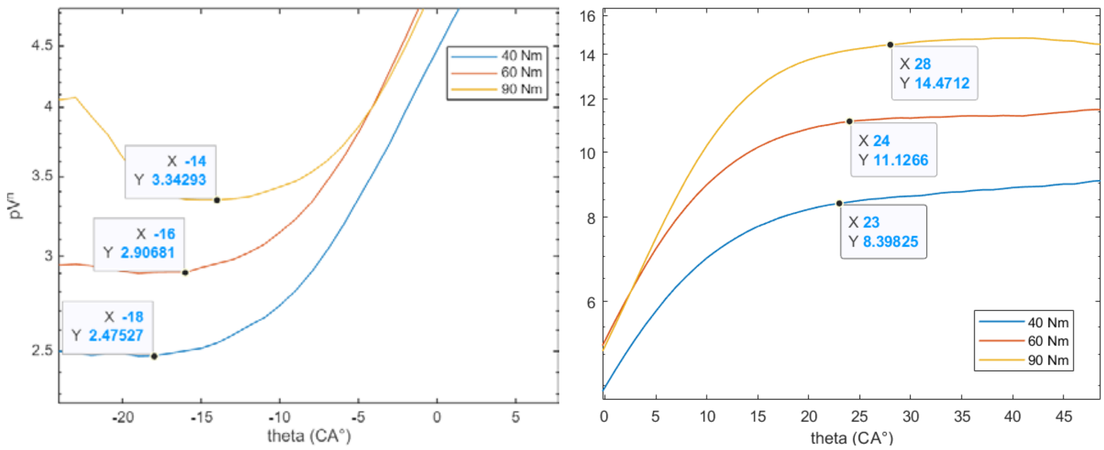


Figure 1: SOC (on the left-hand side) and EOC (on the right-hand side) identification on pV^n diagram (N=2400 rpm)

Mfb is evaluated at any crank angle during the combustion process using Rassweiler-Withrow method:

$$m_b = \frac{(pV^n)_\theta - (pV^n)_{\text{start}}}{(pV^n)_{\text{end}} - (pV^n)_{\text{start}}}$$

Equation 1: Rassweiler and Withrow mass fraction burned formulation

Selecting an appropriate value for *n* is one of difficulty in applying this pressure analysis procedure: heat transfer effects are included only to the extent that the polytropic exponent *n* appropriately differs from γ, as discussed in Appendix A. (Heywood, 2018).

Mfb is crucial for characterizing and optimizing the combustion process, because various combustion-related stages can be derived from it:

- Flame-development period $\Delta\theta_d$ (up to 10% of Mfb).
- Rapid-burning angle $\Delta\theta_b$ (from 10% up to 90% of *Mfb).*
- *Mfb50,* which is usually referred to as barycentre of combustion.

Performing the analysis with fixed speed conditions, it can be observed from Figure 2 that if the load is increased, the spark timing can be retarded allowing to keep the peak firing pressure (PFP) at similar crank angles for the different curves. This is only true for SI engine: in CI engines, if the load is increased, more time for mixing is required and the injection must be advanced consequentially.

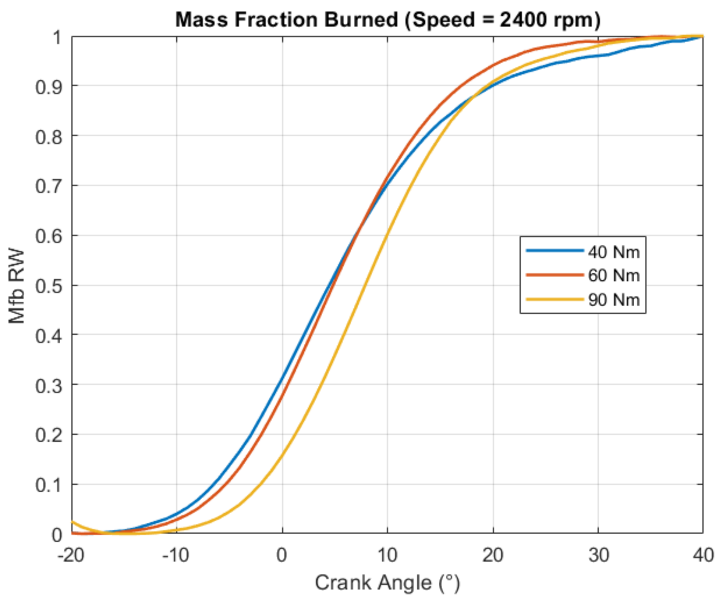


Figure 2: Mass fraction burned for fixed speed

At constant loads, as engine speed increases, time available for combustion decreases with a trend less than linear. To compensate for the longer crank-angle duration needed, combustion is advanced according to:

$$\theta_{comb}[°CA] = 6N[rpm] \cdot t[s]$$

Equation 2: combustion duration depending on engine speed

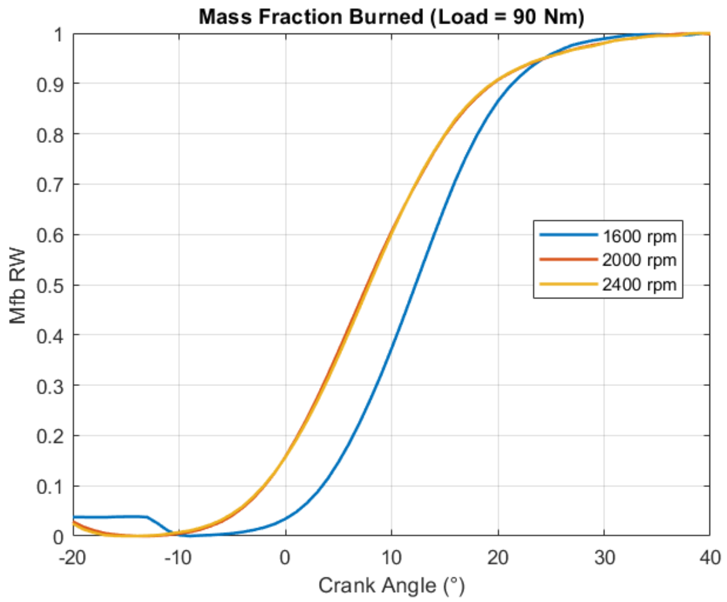


Figure 3: Mass fraction burned for fixed load

HRR and Q_{cum} are estimated using the first law of thermodynamics:

$$HRR = \frac{dQ_{ch}}{d\theta} - \frac{dQ_{ht}}{d\theta} = \frac{1}{\gamma - 1} V \frac{dp}{d\theta} + \frac{\gamma}{\gamma - 1} p \frac{dV}{d\theta}$$

Equation 3: 1st thermodynamics law

Heat release evaluation helps to visualize how pressure rise due to combustion is proportional to the amount of chemical energy released rather than the mass of the mixture burned. Being Q_{cum} the cumulative heat released, it is evaluated on MATLAB applying *cumtrapz*$\left(\frac{dQ}{d\theta}\right)$.

For fixed speed, as the load increases also the peak value of $\frac{dQ}{d\theta}$ increases: it occurs slightly later in terms of crank angles, indicating a stronger combustion phase. Also, Q_{cum} increases with load, reaching higher final values.

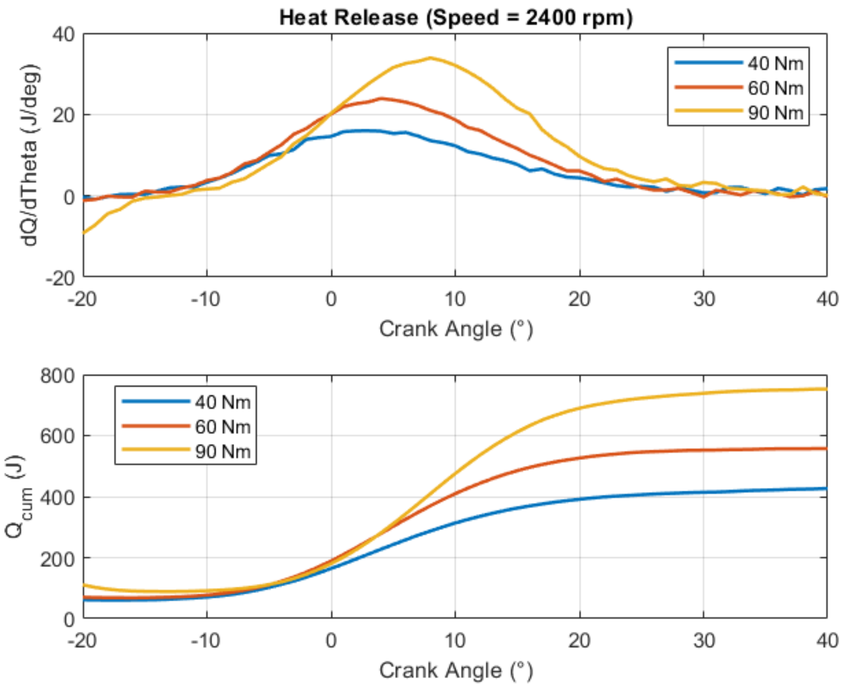


Figure 4: HRR and Q_{cum} for fixed speed

Increasing the speed while keeping the load constant, the peak of heat release decreases and it is slightly anticipated, being closer to the TDC. Additionally, a similar amount of Q_{cum} is reached at the end of the process.

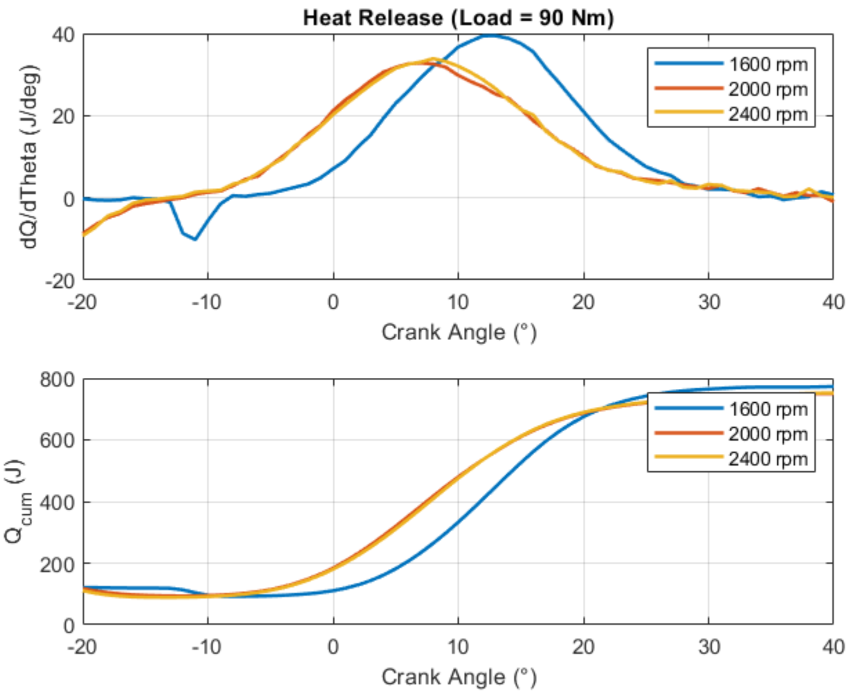


Figure 5: HRR and Q_{cum} for fixed load

By normalizing the cumulative heat release, the dependence on total energy is removed, just as the mass fraction burned curves eliminate the dependence on the amount of fuel. What remains in both cases is the shape of the combustion process. This is why the normalized heat release curve closely matches the Rassweiler-Withrow curve: both represent how combustion evolves as a function of crank angle.

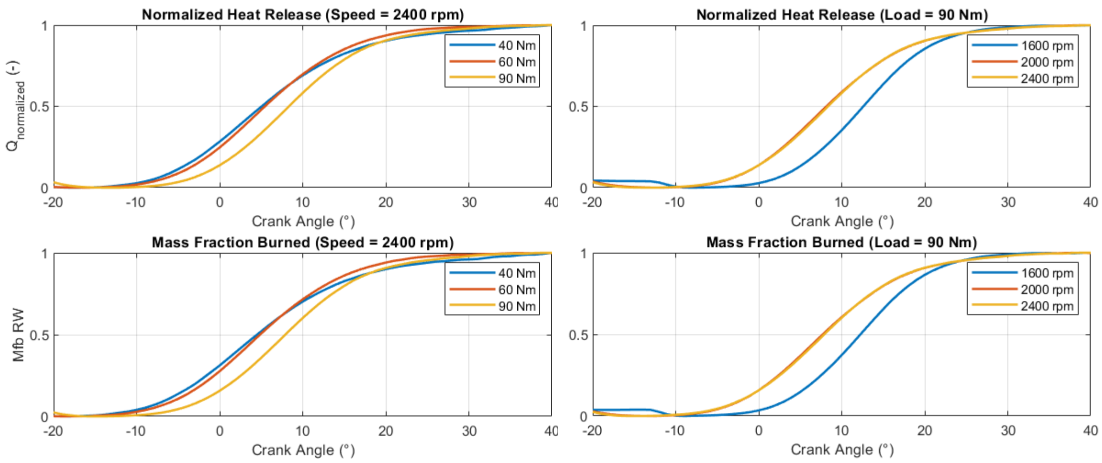


Figure 6: Q_{norm} (on top) and Rassweiler-Withrow curve (below) comparison for fixed speed (left) and fixed load (right).

For one of the operating conditions previously evaluated (1600rpm, 40Nm), a mean of 100 cycles pressure measurements was obtained from *CycleVariation* block in AVL IndiCom and it was then used as input profile to evaluate Q_{cum} (defined as *IntMean* outside the *Thermodynamics* block in Figure 7).

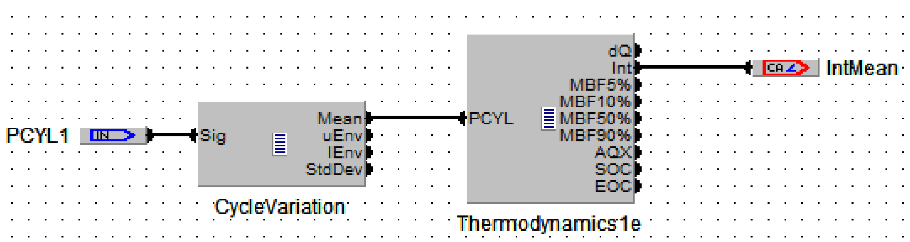


Figure 7: Calgraph application for combustion analysis

As a validation for the performed analysis, it can be observed that the heat release profile extracted from AVL IndiCom closely matches the results carried out on MATLAB, as shown in Figure 8. Differences are mainly related to different input provided (100 cycles for AVL and one cycle for MATLAB).

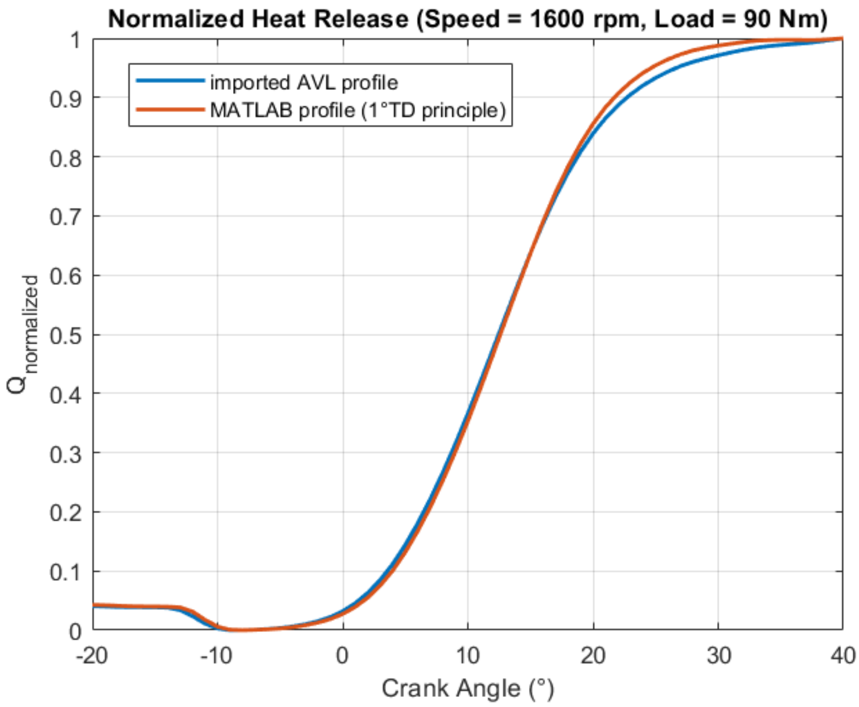


Figure 8: validation of results between MATLAB and AVL Indicom software

2 Advanced combustion analysis – Part II

The investigated area concerns an algorithm realized by *Colin G., Giansetti P., Chamaillard Y. and Highelin P.* for in-cylinder fresh charge estimation using measured cylinder pressure.

Despite being published in 2007, their technical paper remains highly relevant as more stringent pollutant emission regulations highlight the need for engine control strategies to compute the air mass induced into the combustion chamber: even though it is not a measurable entity, it can be estimated on a cycle-by-cycle basis during compression if cylinder pressure is measured.

2.1 Algorithm implementation

The aim of this analysis is to perform an evaluation of the in-cylinder mass on a Spark Ignition turbocharged BMW-Mini engine (whose specifications are shown in Figure 1), validating the accuracy of the estimation through the data gathered from the engine test bed.

While performing the analysis, the following assumptions are introduced:

i. Gases are considered ideal.
ii. In-cylinder mass is considered constant as long as the valves are closed (blow-by is neglected).
iii. Engine is run at equivalence ratio Φ = 1.
iv. Overlap valve is fixed.
v. Fresh air warming up through the runners is neglected.

The steps of the algorithm, shown in Figure 9, were performed based on three operating conditions, analysing how load and speed can affect the residual gas fraction (since the engine is endowed with an EGR valve) and, consequently, the in-cylinder air mass.

	N [rpm]	**Load** [Nm]
cycle n°1	1600	40
cycle n°2	2400	40
cycle n°3	2400	90

Table 3: engine operating conditions evaluated

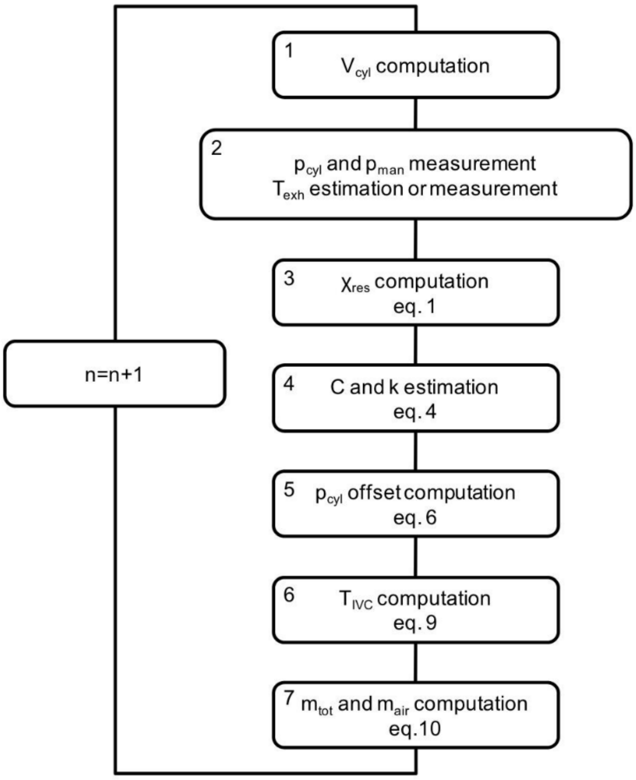


Figure 9: algorithm steps (Colin et al., 2007)

1. Cylinder volume is computed for each crank angle, knowing the engine geometry:

$$V_{Cyl} = \frac{V_d}{r_c - 1} + \frac{V_d}{2} \cdot \left[R + 1 - cos\,\vartheta - (R^2 - sin^2\,\theta)^{\frac{1}{2}}\right]$$

Equation 4

2. Cylinder pressure is recorded at each crank angle; manifold and exhaust pressure are assumed to be coincident with the pressure values at BDC, measured before compression while the intake valve is open at -180°*CA* and after expansion while the exhaust valve is open at 180°*CA* respectively.

	p_{man} [bar]	p_{exh} [bar]
1600-40	1	1,08
2400-40	1,01	1,45
2400-90	1,03	1,62

Table 4: manifold and exhaust pressure for each operating condition

3. The algorithm is based on physical equations, involving a residual gas model: the residual gas fraction is derived from a model calibrated on single-cylinder engine experiments, requiring p_{man}, p_{exh} , Γ and N as input.

$$\chi_{\text{res}} = \frac{2.095}{N} + \frac{\Gamma-1}{\Gamma} \cdot \left(\frac{p_{man}}{p_{exh}}\right)^{0.285} \sqrt{|p_{exh} - p_{man}|} + 0.436\, \frac{\phi^{0.295}}{\Gamma} \left(\frac{p_{man}}{p_{exh}}\right)^{0.719}$$

Equation 5: empirical residual gas formulation

For this application, a more reliable residual gas fraction map was available for the tested BMW-Mini engine: χ_{res} can be graphically evaluated from Figure 8, depending on speed and load.

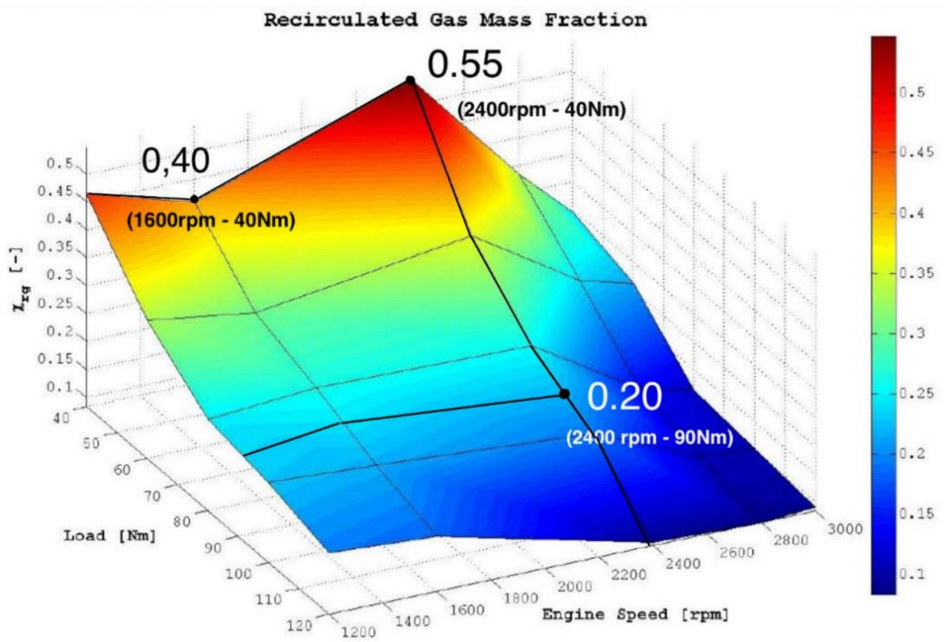


Figure 10: residual gas fraction estimation for three operating conditions from BMW-Mini engine recirculated gas mass fraction map

4. Cylinder pressure and volume are fitted considering a polytropic transformation $\hat{p}_{Cyl} V_{cyl}^{k} = C$ (with constant coefficients *k* and *C*) during compression stroke, while the valves are closed before combustion: the first data point is located immediately after IVC and the last data point is

located just before the ignition. For the identification of valves opening and closing, a valve lift diagram for this specific engine was used.

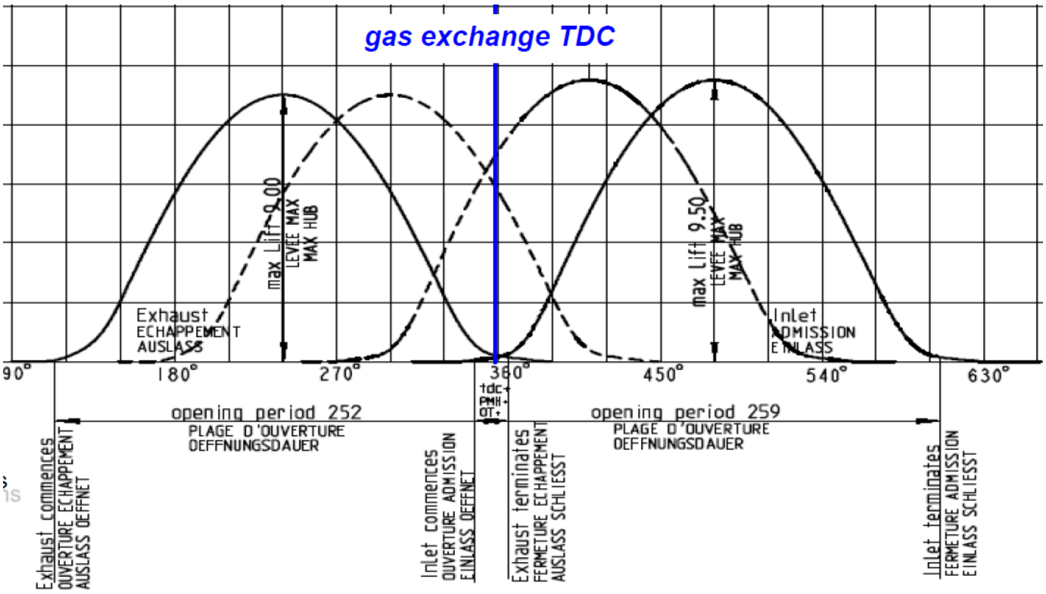


Figure 11: BMW-Mini engine theoretical valve lift diagram

from the diagram		**assuming combustion TDC at 0°**	**unit**
IVO	346	346	°CA
IVC	605	-115	°CA
EVO	115	115	°CA
EVC	367	353	°CA

Table 5: BMW-Mini engine valves opening and closure

The location of the spark for each considered cycle was identified from pressure measurements:

	SOI [°CA]
1600-40	-35
2400-40	-39
2400-90	-23

Table 6: ignition crank angle before TDC for each operating condition

It was observed that noises due to valve opening and closure occurring in adjacent cylinders affect the measurement of in-cylinder pressure in proximity of the interval on which the fitting must be performed.

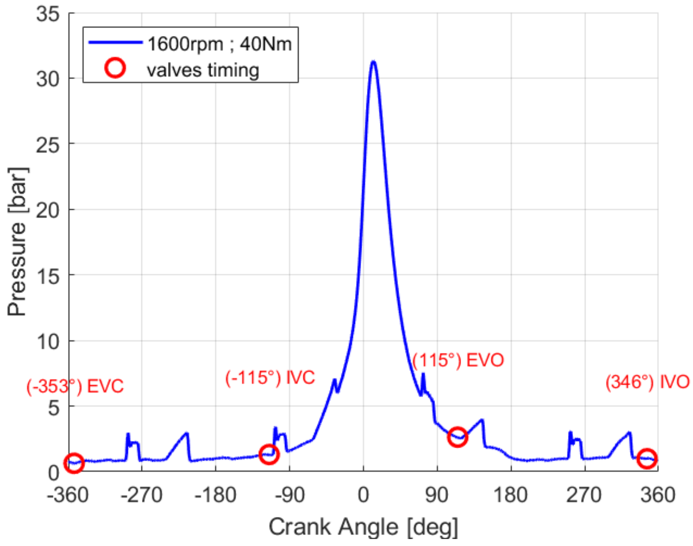


Figure 12: valves timing on in-cylinder pressure measurements

For this reason, a filtering action needs to be locally applied:

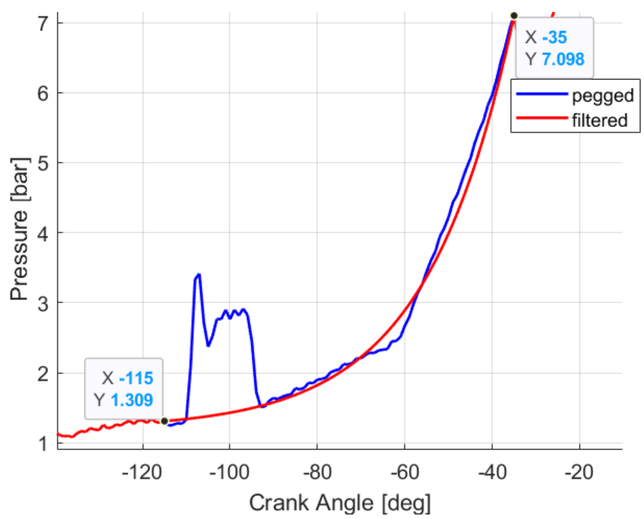


Figure 13: filtered pressure signal from IVC to ignition (1600rpm, 40Nm)

Once the signal is filtered, the values of *k* and *C* are obtained through the least square method on MATLAB, as shown in Appendix B.

$$X = (H^t H)^{-1} \cdot H^t Y$$

$$X = \begin{bmatrix} k \\ \ln C \end{bmatrix}, H = \begin{bmatrix} -\ln V_1 & 1 \\ \vdots & \vdots \\ -\ln V_i & 1 \\ \vdots & \vdots \\ -\ln V_n & 1 \end{bmatrix} \quad Y = \begin{bmatrix} \ln p_1 \\ \vdots \\ \ln p_i \\ \vdots \\ \ln p_n \end{bmatrix}$$

Figure 14: least square method vectors, fitted from IVC to ignition

	k	C [Pa*m^3]
1600-40	1,236	6,356
2400-40	1,223	8,757
2400-90	1,391	2,165

Table 7: fitted polytropic coefficient

The in-cylinder mass $\hat{m}_{Cyl}$ can be computed if the ideal gas constant of the in-cylinder mixture and in-cylinder temperature (evaluated at IVC, when in-cylinder mass is fixed) are known.

5. r_{cyl} is computed from ideal gas constant of fresh charge ($r_{man} = 287 \frac{J}{k_g \,°K}$) and residual gases ($r_{exh} = 289 \frac{J}{k_g \,°K}$) accounting for the residual gas fraction χ_{res} of each condition.

$$r_{cyl} = (1\text{-}\chi_{\text{res}})r_{man} + \chi_{\text{res}}r_{exh}$$

Equation 6: weighted ideal gas constant of the in-cylinder mixture

	χ_{res}	r_{cyl} [J/kg°K]
1600-40	0,40	287,8
2400-40	0,55	288,1
2400-90	0,20	287,4

Table 8: data for ideal gas constant evaluation

6. T_{IVC} is evaluated using the first law of thermodynamics, considering identical specific heat capacities and approximating $\Delta T_{h_t} = 0$ (from assumption v).

$$T_{IVC} = (1\text{-}\chi_{\text{res}})\, T_{man} + \chi_{\text{res}}T_{exh} + \Delta T_{h_t}$$

Equation 7: weighted in-cylinder temperature evaluated at IVC

Dataset for each operating condition analysed includes T_{man} and T_{exh} as measured data.

	χ_{res}	T_{man} [°K]	T_{exh} [°K]	T_{IVC} [°K]
1600-40	0,40	300,1	762,1	484,9
2400-40	0,55	312,1	871	619,5
2400-90	0,20	307	899,6	425,5

Table 9: data for in-cylinder temperature evaluation at IVC

7. Using the ideal gas law (as mentioned in assumption i), $\widehat{m}_{Cyl}$ and $\widehat{m}_{air}$ can be finally evaluated:

$$\widehat{m}_{Cyl} = \frac{C\ V_{IVC}^{1-k}}{r_{cyl}\ T_{IVC}}$$

Equation 8: in-cylinder mass

$$\widehat{m}_{air}(\mathrm{n}) = \widehat{m}_{Cyl}(\mathrm{n})(1\text{-}\chi_{\mathrm{res}}(n))$$

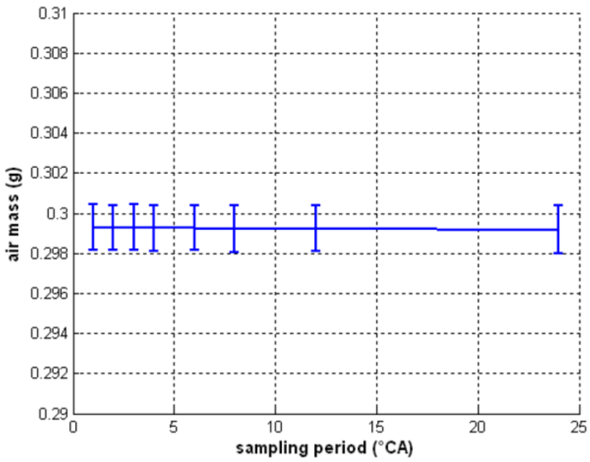
Equation 9: in-cylinder air mass

	χ_{res}	k	C [Pa*m^3]	r_{cyl} [J/kg°K]	T_{IVC} [°K]	m_{cyl} [kg]	m_{air} [kg]
1600-40	0,40	1,236	6,356	299,712	484,9	2,97E-04	1,78E-04
2400-40	0,55	1,223	8,757	304,514	619,5	2,89E-04	1,30E-04
2400-90	0,20	1,391	2,165	291,33	425,5	3,98E-04	3,18E-04

Table 10: in-cylinder fresh air mass estimation for each operating condition

2.2 Results validation

As a preliminary validation, it can be noticed that the results analysis performed in the reference paper (Colin et al., 2007) refers to a fresh admitted mass with the same order of magnitude as the one computed in Table 10, with evidence supported through Figure 15.



Number of cylinder	1
Cycle	4-stroke
Number of intake valve	2
Number of exhaust valve	2
Bore	88.00 mm (3.45 in)
Stroke	82.00 mm (3.23 in)
Combustion chamber	Pent roof
Displacement	498.7 cm^3
Connecting rod length	138 mm (5.39 in)
Piston	Plate
IVO IVC	0 to 48 before TDC 96 to 48 After BDC
EVO EVC	109 to 61 before BDC 0 to 48 After TDC
Compression ratio	10.1:1
Max Valve lift (In/Ex)	9.6/9.4 mm
Valves diameter (In/Ex)	29.5/26.6 mm

Figure 15: air mass estimation with several sampling periods for an engine with geometric characteristics listed beside. N=2000rpm, p_{man}=600mbar (Colin et al.,2007)

Nevertheless, in deeper validation is needed to be performed for the BMW-Mini engine, whoose geometric characteristics and tested operating conditions are different from the ones analysed from the technical paper on a different engine.

Formulations involving specific fuel consumption and brake power allows to estimate fuel and air flow rate, requiring the actual air/fuel ratio to be known:

$$\dot{m}_f = \frac{bsfc \cdot bp}{3600 \cdot n^{\circ}cyl} \longrightarrow \dot{m}_{air} = {}^{A}/_{F}{}_{act} \cdot \dot{m}_f \longrightarrow m_{air} = \frac{\dot{m}_{air}}{\frac{N}{60} \cdot \frac{1}{2}}$$

Equation 20: procedure for air mass computation from power and consumption data

From engine testing dataset all the data needed for this validation were gathered, including $bsfc$ and bp, while ${}^{A}/_{F}{}_{act}$ was fixed at 14.7 being the engine operated in stechiometric conditions (as mentioned in assumption iii).

	bsfc [kg/kWh]	**bp** [kW]	**m** $_{f\_dot}$ [kg/s]	**m** $_{air\_dot}$ [kg/s]	**m** $_{air}$ [kg]	**m** $_{air}$ [kg]	**accuracy**
1600-40	0,321	6,7	1,49E-04	2,20E-03	1,65E-04	1,78E-04	91,7%
2400-40	0,310	10	2,15E-04	3,16E-03	1,58E-04	1,30E-04	82,1%
2400-90	0,257	22,6	4,03E-04	5,93E-03	2,96E-04	3,18E-04	92,7%

Table 11: in-cylinder air mass computation workflow using data from test results (bsfc and bp), and validation with previously estimated values

The algorithm demonstrates strong predictive capability, suggesting that the correlation implemented from the technical paper is robust as it consistently reproduces the air flow dynamics across different speeds and loads. The relative error between the estimated (blue) and experimental (green) air mass values is moderate, varying from 8 up to 18%.

The largest deviation observed, close to 18% for the second study case, might be related to the multiple assumptions previously mentioned for this computation, and to some critical limitations whose impact cannot be neglected:

1. Being performed and calibrated on a single cylinder engine, the empirical formulation of χ_{res} carried out from the technical paper (Equation 5) was not suitable to be extended for the BMW-Mini engine.
   On the residual map shown in figure 10, the values of χ_{res} are approximated to the closest scaled value on vertical axis due to the limitations of the colour scale ($\pm 0{,}05$ resolution).
2. Valves opening and closure diagram, shown in figure 11, has a grid resolution of 30°CA, making the exact valves timing difficult to identify in a precise way.
3. The estimation is affected by the chosen filtering approaches. This aspect is not considered in the reference paper, since the noise mainly originates from valves and combustion dynamics of adjacent cylinders, while the engine tested by the authors is a single-cylinder one.

2.3 Sensitivity analysis for filtering methodology

Two different noise-removal approaches are compared below, focusing on the interval from IVC to the spark, as previously shown in Figure 13, in order to evaluate their influence on air mass estimation.

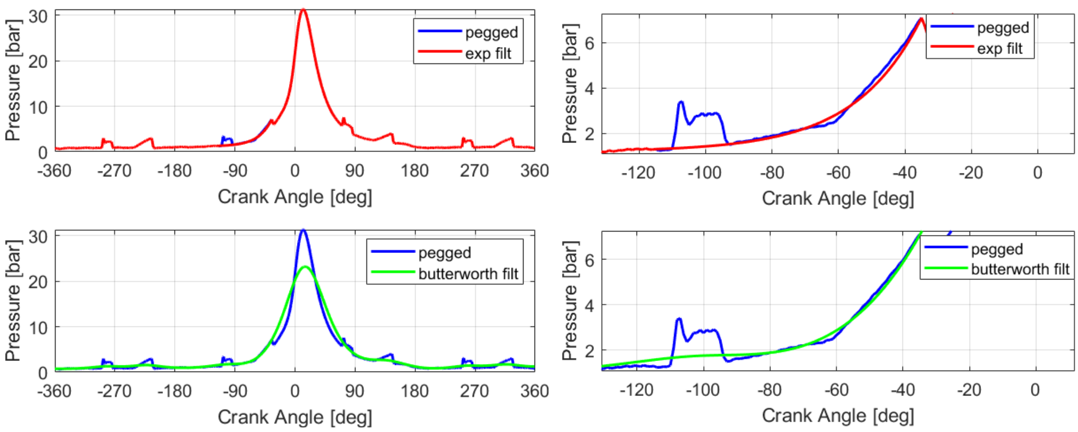


Figure 16: noise removal from IVC to ignition (1600rpm, 40Nm)

Red curve from Figure 16 is not a filter in the classical sense: it is an exponential curve constructed to replace the values between two boundary points, with the control parameter λ which is tuned to control the curvature.

The green profile is a Butterworth filter, whose sampling frequency changes according to the considered engine speed, as documented in Appendix B. Being a low-pass filter, high frequencies are damped: it can be observed from Figure 16 that information about peak pressure is lost if a certain cutoff frequency is applied (D'Ambrosio, 2024). Nevertheless, for the purpose of our application, the focus is solely on the interval from IVC to ignition, during which the occurring noise is filtered.

	exponential accuracy	*butterworth accuracy*	**Cutoff frequency** [Hz]
1600-40	91,7%	92,6%	75
2400-40	82,1%	98,8%	150
2400-90	92,7%	97,8%	200

Figure 17: air mass estimation accuracy carried out with different approaches

Based on the comparison carried out, more accurate results can be obtained using Butterworth filter, provided that the cutoff frequency is properly set so that the filtered profile closely matches the original pressure profile during compression after IVC: as a limitation of this approach, a poorly calibrated cutoff frequency can be the cause of a halved accuracy.

3 Individual reflections

The analysis provided a valuable opportunity to practically understand the power of in-cylinder pressure information, by evaluating quantities, such as the in-cylinder mass, that can be derived from its measurement.

It provided a framework to strengthen my technical background by using AVL for data acquisition and validation, other than MATLAB for data processing, getting acquainted with procedures for in-cylinder mass estimation and manipulation of pressure profiles.

As a further improvement for this analysis, a calibration of the parameters from equation 5 for the BMW-Mini engine could lead to an adaptable residual gas model and, consequently, more accurate results.

(Words count: about 2050 – caption excluded)

References

Colin, G. et al. (2007) “In-Cylinder Mass Estimation Using Cylinder Pressure,” SAE Technical Paper [Preprint]. Available at: https://doi.org/10.4271/2007-24-0049.

D’Ambrosio, S. (2024) Engine testing and HRR analysis*: Propulsion Systems and their applications to vehicles* slides. Politecnico di Torino, Automotive Engineering.

Heywood, J.B. (2018) Internal combustion engine fundamentals. Second edition. New York: McGraw-Hill Education

Definitions, acronyms and abbreviations

C polytropic constant

k polytropic exponent

m mass (kg)

$\dot{m}_{air}$ air flow rate $(\frac{kg}{s})$

$\dot{m}_f$ fuel flow rate $(\frac{kg}{s})$

N engine speed $(\frac{rev}{min})$

p pressure (bar)

r ideal gas constant $(\frac{J}{kg\ °K})$

r_c compression ratio (-)

T temperature (K)

V volume (m^3)

ϑ crank angle (°CA)

χ_{res} gas mass fraction

Subscripts

cyl: cylinder

d: displacement

exh: exhaust

man: manifold

res: residual

Acronyms

$^{A}/_{F_{act}}$ Actual air fuel ratio

BDC: Bottom Dead Center

bp brake power (kW)

$bsfc$ brake specific fuel consumption (kg/kWh)

EOC End of combustion (°CA)

EVC Exhaust Valve Closure (°CA)

EVO Exhaust Valve Opening (°CA)

IVC Intake Valve Closure (°CA)

IVO Intake Valve Opening (°CA)

Mfb Mass Fraction Burned

SOC Start of combustion (°CA)

TDC: Top Dead Center

Appendix A

γ	Specific heat ratio	1.35
n	polytropic exponent	1.3

The specific heat ratio γ of both burned and unburned gases decreases as temperature rises; however, using a constant γ during combustion proves to be a sufficiently accurate simplification (Heywood, 2018).

The exponent *n* for compression and expansion is typically 1.3 (±0.05) for most fuels. This value is close to the average γ_u of the unburned mixture during compression, but it exceeds γ_b for the burned gases during expansion because of heat losses to the combustion chamber walls (Heywood, 2018).

Appendix B

Exponential interpolation (case 1600 rpm – 40 Nm)

```
theta1 = -115;                 %from IVC (or even before IVC)
theta2 = -35;                  % up to the spark
```

```
idx1 = find(theta == theta1);
idx2 = find(theta == theta2);
p_exp = p_cyl_mean;           % Copy of signal
y1 = p_cyl_mean(idx1);        % Boundary values to correct
y2 = p_cyl_mean(idx2);        % Boundary values to correct
n = idx2 - idx1 + 1;          % Amount of points to be corrected
lambda = 4;                   % Control parameter
x = linspace(0,1,n)';
sig_curve = y1 + (y2 - y1) * ((exp(lambda * x) - 1) / (exp(lambda) - 1));
sig_curve(1) = y1;
sig_curve(end) = y2;
p_exp(idx1:idx2) = sig_curve; % Substitution of the interval
```

Butterworth filter (case 1600 rpm – 40 Nm)

```
fs = N/60*360;                %sampling frequency
fc = 75;                      % cutoff frequency
Wn = fc/(fs/2);               % ratio between cutoff and Nyquist freq
n = 2;                        % order of the filter
[b,a] = butter(n,Wn);
p_butter = filtfilt(b, a, p_cyl_mean);
```

Least square method (applied from IVC to ignition)

```
V = V_cyl(IVC:ignition,:);        %m^3
p = p_exp(IVC:ignition,:)*10^5;   %Pa
H = [-log(V), ones(size(V))];
Y = log(p);
X = (H' * H) \ (H' * Y);      %lsq method
k = X(1);
lnC = X(2); C = exp(lnC);
```

ENGR7029 –Advanced Combustion analysis Course work

Course Work Title	**Advanced Combustion Analysis**	
Student Number		**19387525**
Student Name	**Guiseppe**	
Mark %	**Part I (30%)**	**76**
	Part II (70%)	**79**
	Total %	**78**

Turnitin Similarity **27% (includes 19% from the Cover page)-No concern**

Evidence files Excel spreads with input data and the analysis, 4 MATLAB script files used for data analysis part 1 and Part 2, a pdf showing mini engine CAM data, Draft proposal showing detailed methodology developed for implementation and copy of Colin et al (2007) paper.

ENGR7029 Advanced Combustion Analysis- Part I

		Remarks	Marks %
Analysis	Mass fraction burned: Methods used for deriving mass fraction burned, effect of operating conditions, application of mass fraction burned for engine control and optimization (40%)	General principle for deriving mass fraction burned values using cylinder pressure is explained. The results shown in this report show excellent level of subject understanding. The critical reasoning given for the shape of the curve is supported by the literature. The application of this method for engine control and optimization is stated for spark timing but not extensively,	75
	Combustion and Heat release: Methods used for deriving mass fraction burned, effect of operating conditions, relationship between mass fraction burned and heat release and application of mass fraction burned for engine control and optimization (40%)	Similar to mass fraction burned the general principle for deriving rate of heat release and cumulative heat release values using cylinder pressure is explained. The results shown in this report show excellent level of subject understanding. The critical reasoning given for the shape of the curve is supported by the literature. Normalised heat cumulative heat release estimated using the data is compared with estimation obtained from AVL system for checking the calculation procedure.	75
Report	Quality of presentation- format of presentation, quality of tables and figures, reference citation (20%)	Overall, it is a very well presented report. The figures convey the purpose very clearly with all essential details and they reflect excellent level of subject understanding. Appropriately cited and follows standard citation scheme well.	78
		Total	**75.6**

ENGR7029 Advanced Combustion Analysis- Part II

Assessment scheme based on AHEP4 Learning Outcomes

	AHEP Learning Outcomes	Justification	% Marks
M1	Apply a comprehensive knowledge of mathematics, statistics, natural science and engineering principles to the solution of complex problems. Much of the knowledge will be at the forefront of the particular subject of study and informed by a critical awareness of new developments and the wider context of engineering.	This section evaluates the concept proposed by Colin et al (2007) for in-cylinder fresh charge estimation using measured cylinder pressure. The critical review of the paper and stating the assumptions required for evaluating the concepts demonstrates very good level of subject understanding.	75
M4	Select and critically evaluate technical literature and other sources of information to solve complex problems (40%)		
M2	Formulate and analyse complex problems to reach substantiated conclusions. This will involve evaluating available data using first principles of mathematics, statistics, natural science and engineering principles, and using engineering judgment to work with information that may be uncertain or incomplete, discussing the limitations of the techniques	This report show evidence for developing a systematic approach for evaluating and implementing the scheme proposed in the paper for estimating trapped mass in the cylinder. This report shows evidence for collecting data from evaluating the concept. It also examines the limitations of the proposed approach. Overall, this section demonstrates excellent level of subject understanding for evaluating concepts given in the research paper and showing evidence for implementation.	85
M3	Select and apply appropriate computational and analytical techniques to model complex problems, discussing the limitations of the techniques employed.		
M12	Practical and workshop skills Use practical laboratory and workshop skills to investigate complex problems (40%)		
	Individual reflection section (10%) ● Propose further work that would offer improvements and enhancements. ● Evaluate personal learning and development in terms of technology/hardware/software/group work (if applicable)	The reflections are drawn from the observations and conclusion of this work and they are meaningful. Practical implementation of the concept and its useful reflected in the report meets one of the main learning outcomes.	75
Report	Quality of presentation- format of presentation, quality of tables and figures, reference citation (10%)	Overall, a very well presented report. The figures convey the purpose well. The details given in the tables are complete and citation follows standard citation scheme well.	75
		Total	79

M1: Science, Mathematics and Engineering Principles, M2: Problem analysis, M3: Analytical tools and Technique, M4: Technical literature, M12: Practical and Workshop Skills